

Sebin Sunny

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Summary

Computer Science graduate who builds AI-powered applications end to end, from prototype to tested, deployed systems. Enjoys owning problems, iterating quickly, and shipping reliable software as part of a collaborative team.

Education

B.Tech in Computer Science and Engineering

Karunya Institute of Technology and Sciences, Coimbatore

CGPA: 7.48 | Graduated Jul 2025

Work Experience

Freelance Software Engineer | AI and Backend Development | June 2026 – present

- Built RAG and multi-agent pipelines in Python and LangGraph, reaching 15% retrieval accuracy on a 1000+ queries.
- Developed automated test suites in Pytest, reaching 80%+ code coverage.

Associate Software Engineering Intern | Iziel Healthcare(Onsite) | May 2025 – August 2025

- Developed backend test automation tools in Python, validating over 50 software workflows and reducing manual testing effort by 35%.

Projects

AI-Powered Symptom Triage Assistant: AI-Powered Symptom Triage Assistant: Built a hybrid Retrieval-Augmented Generation (RAG) triage system over a 1,000-record medical knowledge base, with severity classification and confidence scoring, routing unmatched queries to a Llama model; achieved 90% Top-3 retrieval accuracy and 97% structured-output validity across 200 test queries. Developed FastAPI REST endpoints with Pydantic validation for reliable structured responses and error handling. Implemented red-flag detection and confidence-based routing to classify cases into emergency, doctor consultation, or self-care pathways.

Voice to Clinical Assessment AI Platform: Converted patient voice notes into structured clinical assessments using Whisper, LangGraph, LLMs, FastAPI, and Pydantic. Integrated RAG with MongoDB for patient-history retrieval, achieving 92% Top-3 retrieval accuracy across 1,000+ clinical queries. Implemented deterministic grounding and confidence scoring to detect unsupported clinical values and reduce LLM hallucinations. Developed REST APIs for assessment processing, retrieval, and persistent storage in MongoDB. Enabled clinician review, editing, and treatment planning, with reliability validated through 188+ automated tests across the end-to-end pipeline.

Technical Skills

Programming: Python, Java, SQL, JavaScript, HTML/CSS, Object-Oriented Programming, Programming Concepts.

Software Engineering: Software Development, Software Maintenance, Application Development, Continuous Training, Production Support, Delivery of Projects.

Backend & APIs: REST API Development, Application Programming Interfaces (APIs), API Integration, Backend Automation, Web Applications.

Testing & QA: Testing skills, Test Automation, API Testing, Regression, Debugging.

Data: Data Preprocessing, Data Pipelines, Databases, Data Analysis.

Professional Skills: Knowledge of Finance, Assertiveness, Communication Skills(English), Self Motivation, Adaptability, Team Working, Elearning, Analytical Thinking, Rapid Learning, Mental Health, Perseverance, Innovation, Geography.

Publications

AI-Driven Personalized Health & Nutrition Assistant Using DeepSeek and LLaVA —Journal of Neonatal Surgery, 2025.

Certifications and Courses

Full Stack Development (Tap Academy), June 2025

Data Science with Python (Finlatics), Nov 2024

Data Science (Teachnook), June 2023

Cloud Security (Cisco), May 2023

Programming Essentials in Python (Cisco), June 2022